No Learning Under This Protocol: A Preregistered Validity Audit of a Downsized Frozen-Student Replication of Sherpa's Simulator

Large language models are increasingly used as simulated students: synthetic learners that should mirror real students, used to train tutors and study pedagogy. We report a preregistered, independent validity audit of the student simulator (Sherpa), which conditions a weight-frozen (weights held fixed) language model on six preference archetypes via gating mechanisms (preference filters on teacher messages). Replicating only the student side, we generated 300 episodes (six archetypes, 50 trajectories each, five tutoring turns) under a fixed-strategy teacher and compared behavioral statistics against five thousand real student sequences from the ASSISTments 2009–2010 skill-builder dataset. The simulated students do not learn at all: a correctness probe (re-asking the student to solve the problem after tutoring) is frozen behaviorally (constant across turns) at its baseline value in 297 of 300 trajectories, and the 24 preregistered two-sample tests are degenerate by construction—the simulated metrics collapse to point masses while real students show the classic learning curve, so their p-values add nothing beyond the descriptive mismatch. No behavioral metric separates the six archetypes. Gate decomposition shows nearly half of adaptive-gate rejections are fail-closed judge parse failures rather than preference judgments. A post-hoc positive control—injecting the ground-truth boxed answer (the convention of enclosing the final answer in a `\boxed{}` tag) into the probe's history in 162 of 300 episodes (four of six archetypes)—moves zero of the wrong-at-baseline probes to correct. This partial, post-hoc control cannot separate instruction-following from incapacity: the "solve from scratch" probe instruction may simply tell the student to ignore history. The freeze therefore cannot be attributed to preference gating specifically, and our conclusions are scoped to this downsized protocol. **Keywords:** large language models, preregistration, simulated students, validity audit

1. Introduction

Simulated students—language-model agents designed to behave like human learners—are becoming infrastructure. They train tutoring systems (Xu et al., 2026), pilot instructional interventions, and stand in for human subjects in education research. Their scientific value rests entirely on a validity claim: that the simulator's behavior mirrors real student behavior closely enough for conclusions to transfer. That claim is rarely audited independently. The stakes are concrete: a tutoring policy trained against a simulator that cannot learn optimizes pedagogy signals that would never transfer to real students—compute spent training on an instrument that measures nothing.

Sherpa (Xu et al., 2026) proposes archetype-conditioned simulated students: a frozen language

model whose teaching interactions are filtered through preference gates (enforced in the paper's pipeline by a 27B language-model judge around an 8B teacher), producing six "distinct" student archetypes (for example, attempt-diagnosis and subgoal-decomposition). The paper's evaluation targets the *teacher* it trains—student accuracy gains, MathTutorBench pedagogy scores, and a human preference study—while the student simulator's own validity as a learner goes untested—most basically, whether the simulated students learn like students. Real students exhibit one of the most robust phenomena in cognitive science—the power law of practice, in which error rates decay with opportunity (Newell and Rosenbloom, 1981). A simulator that cannot reproduce it is not a student model in any behaviorally meaningful sense. Our probes re-attempt the *same* problem after tutoring, so the test here is the minimal within-problem version—whether instruction leaves any trace on the next attempt (retention), not transfer across problems; a simulator failing even this bar fails the stronger claim a fortiori.

We preregistered an independent audit of Sherpa's student simulator before generating a single trajectory (preregistration locked 2026-10-08; one logged deviation). Replicating only the student side—frozen Qwen3-1.7B with verbatim gate prompts from the released repository—we generated 300 episodes (6 archetypes $\times$ 50 trajectories $\times$ 5 tutoring turns) under a fixed-strategy teacher built from the paper's own teaching templates, and compared behavioral statistics against 5,000 real (student, skill) sequences from the public ASSISTments 2009–2010 skill-builder dataset (Feng et al., 2009).

Our findings are uniformly negative:

1. **Zero learning.** Probe correctness is frozen at its unaided baseline in 297 of 300 trajectories. Mean correctness moves $0.347 \rightarrow 0.343$ across five tutoring turns. Exactly one trajectory improved—with zero accepted tutoring turns (probe noise; see §5.7). Two got *worse* after accepted tutoring.

2. **Degenerate distributional mismatch.** All 24 preregistered two-sample Kolmogorov–Smirnov tests (4 metrics $\times$ 6 archetypes, Holm-adjusted) reject the null. But the "difference" is degenerate: the simulated distributions collapse to point masses—slopes 0.000 ± 0.000, error persistence 0.833 ± 0.000, hint responsiveness 0.000 ± 0.000 across four archetypes, with only the three mixed trajectories giving the other two any spread—while real students show the classic learning curve ($0.499 \rightarrow 0.688$, mean slope -0.35).

3. **Indistinguishable archetypes.** On every behavioral metric, the six archetypes are identical. The only differences are gate acceptance rates (20–46%) and surface reply lengths.

4. **Gate dysfunction, decomposed.** Of 1,500 teacher turns, 17.2% are rejected by the guidance gate, 24.7% by adaptive-gate *parse failures* (the 1.7B judge cannot emit valid tagged XML after three retries; fail-closed), and 26.3% by reasoned preference rejections; 31.9% are accepted. Nearly half the adaptive rejection mass is judge incompetence, not preference enforcement.

5. **No dose–response.** The correlation between accepted tutoring turns per episode and probe improvement is $r = -0.04$ (95% CI $[-0.15, 0.08]$, $n = 300$). Five accepted turns move the probe about as much as zero—and in 23 episodes where the teacher's boxed final answer was *accepted* into the student history, zero probes improved. Post-hoc checking revealed every one of the 45 leaked boxed answers in the 32 accepted messages was numerically

wrong (0/45 correct), so this null indicts the 1.7B teacher's quality rather than testing whether tutoring content can move probes (§5.5).

This is a negative-validity audit, framed as such: the contribution is not a better simulator but a preregistered demonstration that this simulator design, as implemented, exhibits no student-like learning dynamics whatsoever. The gate judge and teacher are downsized to 1.7B (a CPU constraint) versus the paper's 27B judge and 8B teacher, which explains part of the gate dysfunction—but it cannot explain the frozen probes, because even injected true answers move nothing (§5.6).

2. Related work

LANGUAGE-MODEL SIMULATED STUDENTS. The use of language models as stand-ins for human learners has grown rapidly, but validity evidence has lagged behind adoption. Scarlatos et al. (2026) formally define the student-simulation task and benchmark simulation methods on linguistic, behavioral, and cognitive metrics, finding that prompting-based simulators perform poorly; Yuan et al. (2026) identify a "competence paradox"—broadly capable language models asked to emulate partially knowledgeable learners produce unrealistic error patterns and learning dynamics—and argue for epistemic fidelity over surface realism. Sherpa (Xu et al., 2026) is among the most elaborate recent systems: six preference archetypes enforced by language-model judges (a guidance gate plus an adaptive gate). Its evaluation targets the *teacher*—student accuracy gains, MathTutorBench pedagogy scores, and a human pairwise-preference study—while the student simulator's own behavioral validity *as a learner* is assumed, not tested. Our audit is, to our knowledge, the first *independent, preregistered* test of whether a recent simulated-student system reproduces real student *learning dynamics* against real student data: prior validity work targets static response patterns, not learning over time. We test not whether the simulator is internally coherent, but whether it learns like a student at all.

Validation efforts elsewhere cluster around response patterns rather than learning dynamics. Lu and Wang (2024) build knowledge-component-grounded "generative students" whose simulated responses correlate highly with real students' item difficulties—a positive validity result, but for static response patterns, not learning over time. Srivatsa et al. (2025) place eleven language models on the same IRT ability scale as real student populations and find systematic misalignment: strong models overshoot every grade level. An et al. (2026) identify the mechanism: prompt-based simulators answer at their built-in capability level (96.8–100% accuracy) regardless of the assigned mastery profile. Scarlatos et al. (2025) bring knowledge tracing into tutor–student dialogues with language models. These simulators are deployed as training environments for tutors (Xu et al., 2026), as proxy students for piloting assessments (Srivatsa et al., 2025), and as stand-ins for real respondents in item evaluation (Lu and Wang, 2024)—all roles that presuppose student-like behavior. None test whether a simulator's behavior *changes* like a student's under instruction: the learning-dynamics gap our audit targets.

STUDENT MODELING AND LEARNING CURVES. The power law of practice—error rate decaying as a power function of opportunities—is among the most replicated findings in cognitive science (Newell and Rosenbloom, 1981) and underpins knowledge-tracing models from Bayesian knowledge tracing (Corbett and Anderson, 1994) onward. The ASSISTments datasets (Feng

et al., 2009) have anchored decades of student-modeling research. Any claim to simulate students inherits this literature as its validity baseline.

AUDITS AND REPLICATION. Independent audits of high-profile machine-learning artifacts have repeatedly surfaced validity gaps, and preregistered replication is the mechanism by which those gaps become credible rather than anecdotal (Nosek et al., 2018). We adopt that mechanism here.

3. PREREGISTRATION

The full preregistration (locked 2026-10-08, before any study trajectory was generated) is archived with the reproduction package. In brief:

STUDY QUESTION. Do Sherpa's archetype-conditioned simulated students reproduce real student learning dynamics? In a validity audit the roles reverse: the hypothesis of interest is mismatch, and the preregistered tests are a decision-rule formality—the evidential weight lies in the descriptive degeneracy. H1: simulated students differ measurably from real students on at least one of (M1) learning-curve shape, (M2) error persistence, (M3) hint responsiveness. H0: no significant distributional difference on any preregistered metric (24 two-sample KS tests, Holm–Bonferroni, $\alpha = 0.05$).

PROTOCOL. Six preference archetypes—attempt-diagnosis (Att.), subgoal-decomposition (Sub.), contrastive-comparison (Ctr.), causal-justification (Cau.), step-demonstration (Stp.), independent-verification (Ver.)—implemented with frozen Qwen3-1.7B, a "you are Sam, a student" system message, and the verbatim adaptive-gate preference text from the paper's Appendix D.1.2). Per episode: an unaided baseline probe (c_0), then five turns of fixed-strategy teacher instruction (the paper's Appendix D.5.1 templates, strategy matched to the archetype's preference) filtered through the verbatim guidance gate and then the adaptive gate (up to three parse retries, fail-closed, per the released repository), then a greedy probe fork (c_t) graded by the repository's two-stage scorer. Fifty problems per archetype, disjoint, drawn from the paper's shipped filtered MATH test split (528 problems). Real-data comparison: 5,000 seeded (user, skill) sequences from the ASSISTments 2009–2010 skill-builder data (first five opportunities, main problems only).

METRICS. M1: per-trajectory power-law exponent from $\log(e_t) = \alpha + b\log(t)$ on smoothed error rates (constant probe vectors yield slope exactly 0 by construction)—it measures whether the student improves with practice, the learning-curve signature. M2a: Laplace-smoothed $P(\text{err}_{t+1} \mid \text{err}_t)$—whether mistakes persist or get corrected (for constant probe vectors this equals $(4+1)/(4+2) = 0.833$ by the smoothing, a definitional constant rather than an estimated persistence). M2b: maximum error streak—the depth of uncorrected failure runs. M3: for simulated trajectories, mean(c_t |guidance accepted) minus mean(c_t |guidance rejected); for real trajectories, next-opportunity correctness after a hint versus after no hint—whether instruction moves subsequent performance. Full definitions are in the preregistration.

ONE DEVIATION. On 2026-10-09, the context window was raised from 2048 to 4096 tokens after a deterministic overflow at episode 31 crashed every relaunch. Context size does not change

model logits for prompts that fit in 2048 tokens (verified post hoc from the banked artifacts: no single model call in the 31 banked episodes exceeded 1,470 prompt tokens), so the 31 banked episodes remain comparable; the alternative (dropping the episode) would have broken the 300-episode design. The post-hoc answer-injection control (§5.6) is likewise logged in the deviation log as a post-hoc addition: it was stopped after 162 of 300 episodes (2-CPU compute contention at roughly five minutes per episode; episodes processed in file order, so the last two archetypes never ran).

STATED LIMITATIONS. Competition-mathematics versus K-12 domain mismatch; downsized 1.7B judge and teacher (CPU constraint); five turns versus the paper's ten; Q4_K_M quantization (taken as a stated assumption). See §7 for the full discussion.

PREREGISTRATION BOUNDARY. The preregistration covers the 24 KS tests and the metrics in §5.1–§5.3. The gate decomposition (§5.4), boxed-answer and dose–response analyses (§5.5), the probe-stochasticity check (§5.7), and the answer-injection positive control (§5.6) are exploratory, post-hoc analyses conducted after the frozen-probe result was observed; they are labeled as such and carry no preregistered decision rule.

4. METHODS

Protocol at a glance. Probe: a greedy (temperature-0.0) re-attempt of the *same* problem by the frozen student, run on a fork of the accepted-turn history; the unaided baseline probe (c_0) sees no tutoring, each post-tutoring probe (c_1–c_5) sees the full accepted history. **Gate:** the two-stage filter—the guidance gate, then the archetype's adaptive gate—deciding whether a teacher turn enters the student's visible history; rejected turns are hidden and replaced by a scripted complaint. **Fork:** every probe branches off the current accepted history instead of continuing the dialogue, so probe correctness measures whether the tutoring history taught anything.

4.1. SIMULATION

All generation used llama-cpp-python on CPU (Qwen3-1.7B Q4_K_M GGUF). Generation parameters were locked in the preregistration: teacher sampling at temperature 0.7, gates at 0.0, student replies at 0.7, probes greedy (temperature 0.0), intended as deterministic state read-outs—an assumption that is violated: greedy probes are generation-stochastic, though their graded verdicts are stable (see §5.7)—and a temperature-0.0 answer-judge fallback. Each probe is a fork of the current accepted-turn history: the baseline probe (c_0) sees only the system prompt, while each post-tutoring probe (c_t) re-attempts the task with the full accepted tutoring history in context (rejected turns excluded), so tutoring that teaches would move the probe. Thinking mode was disabled in the paper. The master seed was 20261008 with per-episode seeds derived by hashing. Rejected turns never enter the student-visible history (the repository's hidden-from-student turn semantics); the student instead emits the paper's scripted complaint. Every generation is logged with seeds (more than 7,800 generations in total).

FILTER FIDELITY. The shipped filtered split is described as retaining problems the student fails unaided, yet our greedy baseline probes answer 34.7% correctly—which is not a filter failure. Sherpa's own Table 1 reports the student's overall unaided accuracy on these problems as 36%, and our 34.7% $\pm$ 2.7% reproduces it ($z = -0.49$ against 36%). The paper's unaided estimate comes from 8 repeated samples per problem with thinking disabled, matching our greedy no-think probe; the filter retains problems with "room to improve," not guaranteed failures. Baseline behavior is therefore consistent with the paper's own calibration, and the freeze cannot be dismissed as a filter artifact.

4.2. REAL-DATA MATCHING

We used the ASSISTments 2009–2010 skill-builder data (401,756 rows, 4,217 students, 123 skills; officially free to use). Analysis units are (user, skill) sequences with filters for main problems only, tutor mode, at least five opportunities, and complete correctness records; the documented order-identifier de-duplication was applied. The comparison pool is a seeded sample of 5,000 eligible sequences. Domain mismatch is a stated limitation (see Section 3): real trajectories span five opportunities on different problems sharing a skill, while simulated trajectories span five turns on one competition-mathematics problem. The comparison is distributional, not item-matched.

4.3. OUTPUT COMPLETENESS

Before any analysis, we verified output completeness: 300 of 300 episodes present (6×50 exactly), five turns each, all probe vectors complete and binary, 300 unique problems disjoint across archetypes, and all internal consistency checks passing (rejected turns carry no student reply plus a complaint; accepted turns carry replies; correctness vectors match baseline plus probes). No partials and no gaps.

Preregistered exclusions: M2a excludes trajectories with no errors in the pair window (12, 24, 13, 23, 12, 19 per archetype in Att./Sub./Ctr./Cau./Stp./Ver. order; 1,146 of 5,000 real sequences); M3 excludes trajectories with all-accepted or all-rejected turns (8, 3, 29, 21, 5, 6; 3,531 real). The KS table in Appendix A reports effective n per test.

5. RESULTS

5.1. THE SIMULATED STUDENTS DO NOT LEARN

Figure 1 shows mean probe correctness by position. Real students trace the classic learning curve, $0.499 \pm 0.007 \rightarrow 0.688 \pm 0.007$ over five opportunities. Every simulated archetype is flat: pooled correctness moves $0.347 \rightarrow 0.343$. In 297 of 300 trajectories, the probe vector is constant at its baseline value (195 all-wrong, 65.0%; 102 all-correct, 34.0%); the remaining three (1.0%) are mixed. The single improving trajectory had zero accepted tutoring turns. The two trajectories that got *worse* after accepted tutoring (`contrastive-comparison_016`, `causal-justification_044`, both correct at baseline) are the only hint in the data of tutoring-induced harm, a thread for follow-up work rather than a claim.

5.2. DEGENERATE DISTRIBUTIONAL MISMATCH

Table 1 reports the preregistered metrics (mean $\pm$ SD). The 24 KS tests are a preregistered formality: with the simulated distributions collapsed to point masses, the p-values add nothing

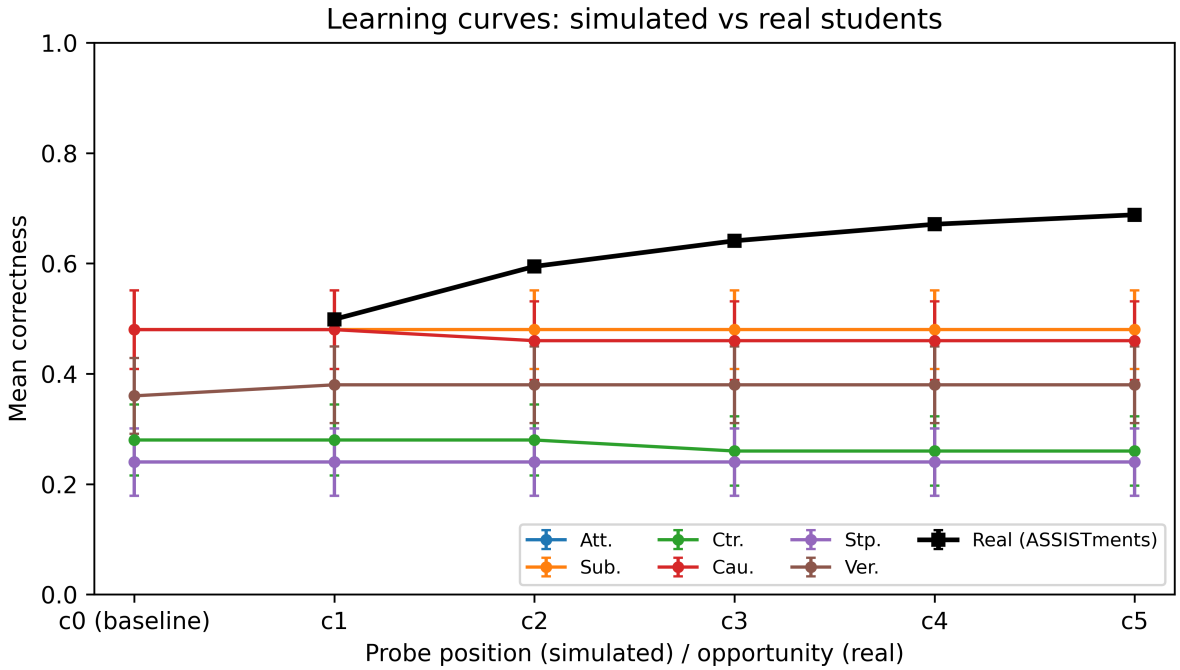


Figure 1: Mean correctness by probe (simulated) or opportunity (real). Real students (ASSISTments, n=5,000) show the power-law learning curve; all six simulated archetypes (n=50 each) are flat. Error bars show $\pm$SE.

beyond the descriptive degeneracy below, which is where the evidential weight lies. The simulated distributions are degenerate: M1 slopes $\equiv 0.000 \pm 0.000$, M2a $\equiv 0.833 \pm 0.000$, and M3 $\equiv 0.000 \pm 0.000$ in four archetypes, the other two showing only trace spread from the three mixed trajectories. Real students show M1 mean slope -0.35 (SD 1.03), M2a 0.467 (SD 0.184), M2b mean streak 1.60 (SD 1.41), and M3 mean hint lift 0.087 (SD 0.498).

All 24 preregistered KS tests reject H0 after Holm correction (the full table, with raw p-values, sample sizes, and bootstrap confidence intervals, is archived with the reproduction package). Per the preregistered decision rule, H1 is supported on all metric families—but the "difference" is degenerate: the simulator does not behave like a *different kind* of student; it behaves like *no student at all*.

CONSTRUCT-VALIDITY NOTE. The preregistered comparison is distributional, not construct-matched: simulated trajectories span five turns on a single competition-mathematics problem, while real trajectories span five opportunities across different problems sharing a skill; the M3 operationalizations likewise differ (accepted-vs-rejected guidance versus hint-vs-no-hint, the latter confounded by help-seeking selection). Rejection was therefore expected even for a competent single-problem simulator. The evidential weight of this section lies not in the p-values but in the descriptive degeneracy documented above: point-mass simulated distributions where real students show spread and learning.

Table 1: Preregistered metrics, mean ± SD. Simulated n=50 per archetype; real n=5,000. Archetype abbreviations: Att.=attempt-diagnosis, Sub.=subgoal-decomposition, Ctr.=contrastive-comparison, Cau.=causal-justification, Stp.=step-demonstration, Ver.=independent-verification. Post-exclusion counts are archived with the reproduction package.

Archetype	M1 slope	M2a persist.	M2b streak	M3 hint-resp.
Att.	0.000±0.000	0.833±0.000	3.80±2.14	0.000±0.000
Sub.	0.000±0.000	0.833±0.000	2.60±2.50	0.000±0.000
Ctr.	0.045±0.312	0.831±0.014	3.66±2.19	−0.024±0.106
Cau.	0.035±0.244	0.832±0.006	2.68±2.48	−0.011±0.061
Stp.	0.000±0.000	0.833±0.000	3.80±2.14	0.000±0.000
Ver.	0.000±0.000	0.833±0.000	3.10±2.43	0.000±0.000
Real	−0.350±1.032	0.467±0.184	1.60±1.41	0.087±0.498

5.3. Archetypes are behaviorally indistinguishable

On M1 through M3 the six archetype distributions coincide (all degenerate at the same values); a classifier given only these metrics could not beat chance at telling archetypes apart. The differences that exist do not constitute distinct learning dynamics: gate acceptance rates of 20–46% (χ^2=54.1, $p \approx 2 \times 10^{-10}$, df=5; 95% Wilson confidence intervals archived; turns nested within episodes, so the p-value is descriptive) and surface reply lengths of 811–1,070 characters (Kruskal–Wallis $p \approx 6 \times 10^{-7}$ with heavy overlap; likewise descriptive), and baseline accuracy of 0.24–0.48 (confounded, because problem sets are disjoint by preregistered design). Sherpa's premise is that preference gates produce *distinct student profiles*; here they produce no learning dynamics at all.

5.4. Exploratory: gate decomposition—half the rejections are judge failures

Of 1,500 teacher turns, gate outcomes decompose as follows.

Outcome	Turns	Share
Rejected by guidance gate	258	17.2%
Rejected by adaptive gate, judge parse failure (fail-closed)	370	24.7%
Rejected by adaptive gate, reasoned preference failure	394	26.3%
Accepted	478	31.9%

370 of 764 adaptive-gate rejections (48.4%) are the 1.7B judge's inability to emit valid tagged XML, not preference judgments (Figure 2). Conditional on valid judge output, the adaptive gate accepts 54.8% of turns. A manual audit of 30 sampled rejections (single rater; no inter-rater reliability measured, so the reasoned-vs-parse split is approximate) found all 15 reasoned rejections defensible (for example, correctly flagging single-alternative messages for the contrastive archetype); the other 15 were empty-reason parse failures. The reasoned-reject rate (26.3%) is the cleaner estimate of genuine gate strictness—and it directly bounds how much of the headline 68.1% rejection rate survives the judge-downsizing caveat.

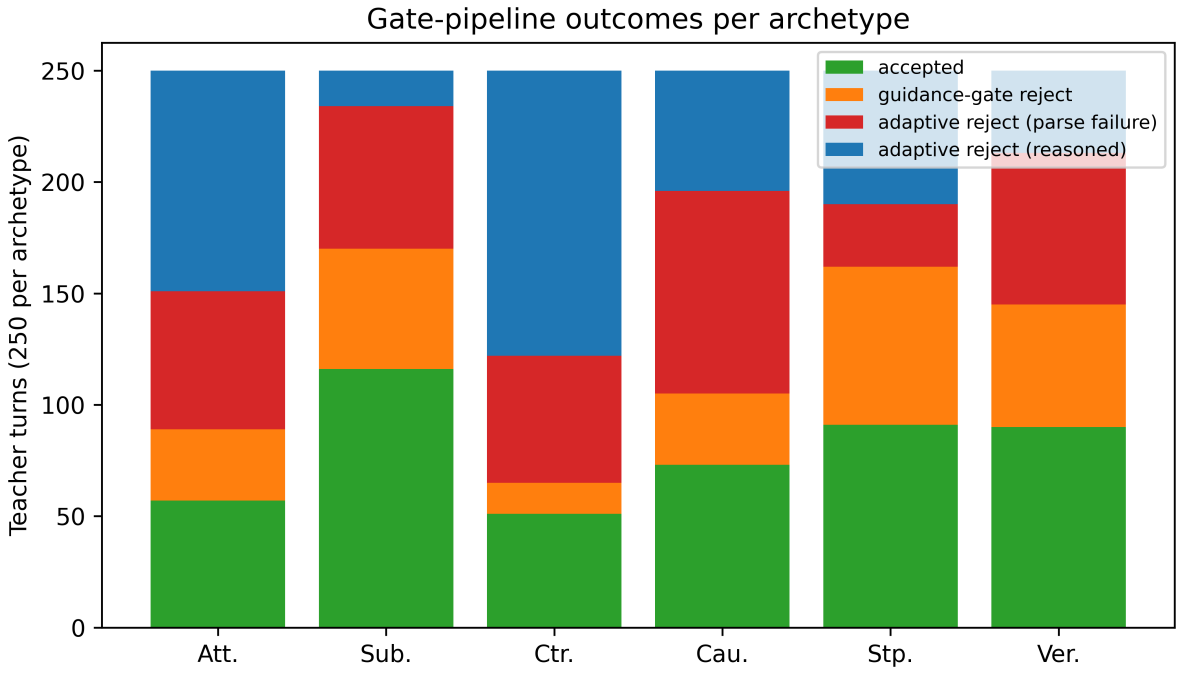


Figure 2: Gate-pipeline outcomes per archetype (250 turns each). Nearly half (370 of 764, 48.4%) of adaptive-gate rejections are fail-closed 1.7B-judge parse failures (red), not preference judgments (blue).

5.5. Exploratory: leaked boxed answers are all wrong; no dose–response

The fixed-strategy teacher was instructed never to reveal the final answer, yet **141 of 1,500 (9.4%) teacher messages contain boxed final answers** (the final answer enclosed in a `\boxed{}` tag)—the 1.7B teacher emits them regardless. The guidance gate caught 80; 29 were adaptive-rejected; **32 were accepted into the student's visible history**. In the 23 episodes containing an accepted boxed answer, **zero probes improved**. Post-hoc checking of the 45 leaked boxed answers inside the 32 accepted messages shows **all 45 were numerically wrong** (0/45 correct against the ground truth): the 1.7B teacher does not merely leak answers, it leaks *wrong* answers (for example, `causal-justification_044` accepted $\boxed{11}$ for a ground truth of 12, then got worse). This null therefore indicts teacher quality rather than testing whether tutoring content can move probes; the clean test is the positive control in §5.6. There is no dose–response either: the correlation between accepted turns per episode and probe improvement is $r = -0.04$ (95% CI $[-0.15, 0.08]$, $n = 300$; Figure 3); five accepted turns move the probe about as much as zero (mean change 0.016, −0.009, −0.016, 0.000, 0.000, 0.000 for zero through five accepted turns; n=62, 106, 63, 40, 19, 10).

5.6. Exploratory: positive control—injected true answers move nothing

The preregistered protocol has no positive control: no configuration in which probes were expected to move, so the freeze could in principle reflect a broken probe loop rather than an unresponsive student. Post hoc—unlike the teacher-leaked (and uniformly wrong) boxed answers in §5.5, here the injected answer is the ground-truth one—we injected the ground-truth boxed answer as an

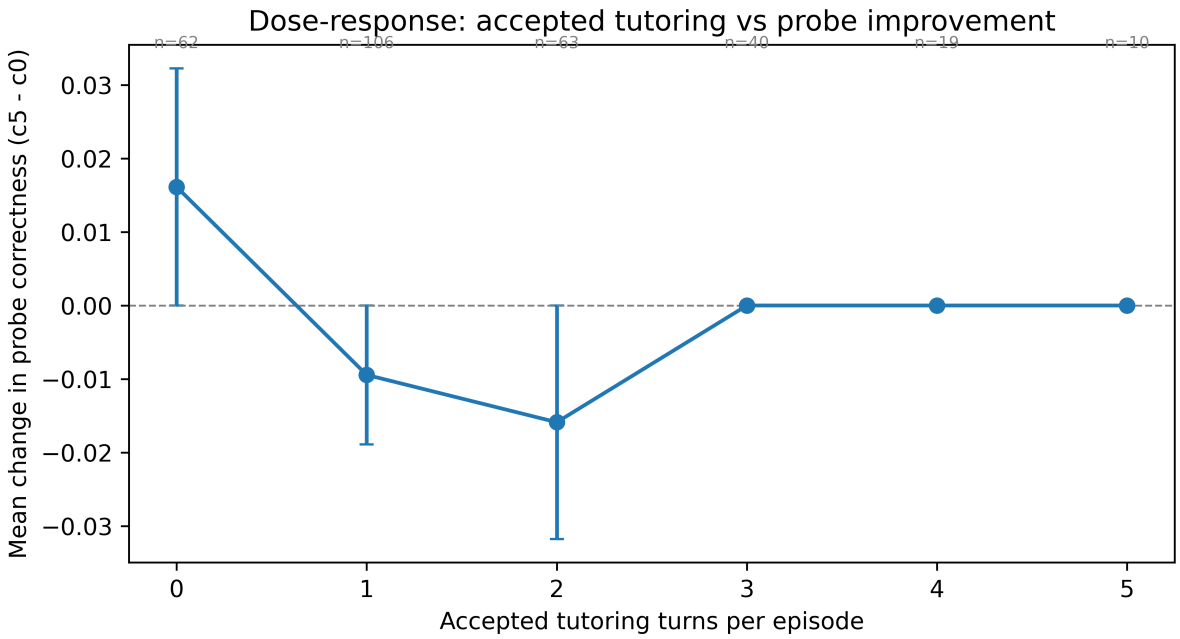


Figure 3: Mean change in probe correctness by accepted tutoring turns per episode. No dose–response ($r = -0.04$). Error bars show ±SE.

accepted teacher turn into the probe history of 162 of the 300 banked episodes and re-ran the identical greedy probe plus two-stage grading (`analysis/answer_injection_control.py`; logged in the deviation log as a post-hoc addition). The control history is built from the injected turn alone: it *replaces*, not augments, the episode's accepted tutoring history, so the injected answer faces no competing history—the more generous test. Of the 106 episodes wrong at baseline, **zero flipped to correct (0/106)**: per archetype, 0/38 attempt-diagnosis, 0/26 subgoal-decomposition, 0/36 contrastive-comparison, 0/6 causal-justification (95% CI [0, 2.8%]; rule of three, $3/106$). Of the 56 already correct, 55 stayed correct and one flipped to wrong (1/56; 95% CI [0.05%, 9.5%], consistent with the probe-noise rate in §5.7). The run stopped at 162 episodes: a 2-CPU VM shared with other jobs ran each episode's probe plus grading in roughly five minutes, and compute contention halted the run. Episodes were processed in file order, so causal-justification stopped at 12 of 50 while step-demonstration and independent-verification (138 episodes combined) were never run.

Two caveats bound what this control shows. First, the probe prompt instructs the student "Now try to solve the problem from scratch:"—a direction that plausibly tells the student to ignore the history above. The 0/106 null is therefore consistent with an instruction-following alternative (the student obeying "from scratch" rather than consulting history) as well as with a genuine inability to use history content; this probe design cannot separate the two, and no absolute "channel is dead" claim is warranted. Second, the control is partial: the 0/106 bound covers the four archetypes run, not the two unrun ones. The probe loop itself is not broken (it answers 34.7% unaided from parametric knowledge), and the control rules out weak tutoring as the sole explanation of the freeze for the four archetypes tested—but the unresponsiveness claim is scoped to this solve-from-scratch probe instruction, not to student capacity in general. The freeze remains an unresponsiveness of the frozen 1.7B student under this probe instruction, not

evidence about preference gating.

5.7. Exploratory: greedy probes are not deterministic read-outs

The preregistration assumed temperature-0.0 probes are deterministic state read-outs. That assumption does not hold: among the 62 trajectories with zero accepted turns (identical probe context all six times), probe *generations* varied in 54 (87%) while graded *verdicts* flipped in only one. Generation-level stochasticity does not propagate to verdicts, so the frozen-correctness finding is not an artifact of re-scoring a single generation; the determinism assumption is dropped.

6. Discussion

What failed. Two candidate explanations fit the preregistered data, which could not adjudicate between them. First, *the tutoring is vacuous*: with most teacher turns rejected and a weak 1.7B fixed-strategy teacher (which leaks wrong boxed answers 9.4% of the time, all of them incorrect), the accepted histories that probes fork from are thin (mean 1.59 accepted turns per episode; 62 episodes with zero). We did not rate the pedagogical quality of the 478 accepted turns, so teaching quality remains an open question. Second, *the student does not use its history*: the post-hoc positive control (§5.6) injects the true boxed answer as the probe history (replacing, not augmenting, the accepted turns) and moves zero of 106 wrong-at-baseline probes (162 episodes, 95% CI [0, 2.8%]). This rules out weak tutoring as the sole explanation, but it does not establish general incapacity: the probe instruction "Now try to solve the problem from scratch" plausibly tells the student to ignore history, so instruction-following is an explicit alternative to incapacity, and the control covers only four of six archetypes. The first hypothesis is true descriptively; under the solve-from-scratch probe instruction, the student is unresponsive to history content regardless of teaching quality. What the data rules out is the middle position: that these archetype-conditioned simulators, as implemented here, exhibit student-like learning dynamics.

Which findings survive which deviation. The 24.7% parse-failure mass is the component most plausibly explained by the 1.7B judge—a 27B judge should reduce it, but the paper itself reports that its 8B training-time judge "struggles to follow the required format in long contexts" (Xu et al., 2026), so format-following failure is documented at the paper's own judge scale and may be partly intrinsic rather than a pure downsizing artifact. The 26.3% reasoned-reject rate remains the cleaner estimate of genuine gate strictness. The frozen probes, the null dose–response, and the accepted-answer-reveal nulls do not depend on judge quality. The fixed-strategy teacher is a deliberate scope choice as well as a constraint: auditing the *student* simulator requires holding teaching fixed, because an adaptive teacher would confound student-side validity with teacher quality. This is also the largest deviation from the paper's own pipeline: its reported 20.5 percentage point improvement in student test accuracy came from an RL-trained 8B teacher with a teacher preparation stage and ten turns (Xu et al., 2026); our fixed-strategy 1.7B teacher, no preparation stage, and five turns deliberately hold none of that constant. The post-hoc positive control (§5.6) narrows the two: bad teaching is real (thin accepted histories, wrong leaked answers), but under this probe instruction the binding constraint is the unresponsive student—even true answers injected into history move nothing. The frozen-probe finding therefore describes the student simulator under this scoped protocol, not Sherpa's full system. The natural

follow-ups are a gates-disabled ablation, a pedagogical-quality rating of accepted turns, and the 27B-judge replication.

WHAT THIS MEANS FOR SIMULATED-STUDENT RESEARCH. The failure mode we document is specific and actionable. Preference gating as a mechanism for *distinct student profiles* is untested by internal metrics: our archetypes differ only in gate strictness and reply length, not in learning behavior. Any evaluation of simulated students should include the cheapest validity check available—a learning curve against a real-student baseline—before more elaborate claims. And gate judges should be evaluated on format reliability, not just judgment quality: nearly half of our adaptive rejections were XML parse failures. The frozen-probe pattern is a concrete instance of the competence paradox (Yuan et al., 2026)—the phenomenon in which a capable model emulating a weaker learner performs at its own capability floor, not the learner's: a broadly capable model asked to emulate a partially knowledgeable learner answers at its built-in capability level (34.7% unaided) and cannot be moved off it by instruction—the "error pattern" is a capability floor, not a student profile.

THE CAVEAT, PLAINLY. The 1.7B downsizing is the sharpest threat to our conclusions, and it cuts unevenly. It plausibly explains *part* of the gate dysfunction (the 24.7% parse-failure mass should shrink under the paper's 27B judge, though the paper documents format-struggle at its own 8B judge scale, so this mass may not dissolve entirely; the 26.3% reasoned-reject mass is the cleaner number). It cannot explain the frozen probes: the positive control shows the student does not use even true history content under the solve-from-scratch probe instruction—a scoped unresponsiveness of the weight-frozen 1.7B student under this protocol (the control cannot separate instruction-following from incapacity) rather than a gate effect. (The accepted-answer-reveal null is confounded by the teacher leaking wrong answers, §5.5; the control supersedes it.) A 27B-judge replication is the obvious follow-up; we state exactly what we expect it to change (gate acceptance up) and not to change (flat learning curves).

7. LIMITATIONS

Domain mismatch (competition mathematics versus K-12 skill builder; preregistered); downsized 1.7B judge and teacher (the caveat above); disjoint problem sets confound archetype baseline comparisons; probe stochasticity is unmodeled (though verdict-stable); the language-model answer judge's errors would propagate silently (no preregistered spot-check); five turns versus the paper's ten; fifty trajectories per archetype (CPU budget). A post-hoc positive control (injecting the true boxed answer into the probe history; §5.6) moved zero of 106 wrong-at-baseline probes (162 episodes), so, under the solve-from-scratch probe instruction, the freeze reflects the student's unresponsiveness to history content—it cannot be attributed to preference gating specifically, and the conclusions are scoped to this frozen-student protocol.

8. CONCLUSION

We preregistered and executed an independent validity audit of Sherpa's archetype-conditioned simulated students. The verdict is negative: 300 episodes, no measurable learning, archetypes indistinguishable on learning-dynamics metrics, and a gating pipeline in which half the rejections

are judge format failures. A post-hoc positive control (162 episodes, four of six archetypes; 0/106 flips, 95% CI [0, 2.8%]) shows the student does not use even true answers under the solve-from-scratch probe instruction—though this confounded control cannot separate instruction-following from incapacity. This verdict is therefore scoped to the downsized frozen-student protocol: it is a finding of unresponsiveness under this probe instruction, not an attribution to preference gating and not a claim about Sherpa's full system. Negative results of this kind—preregistered, fully reproducible, with every number traceable to banked artifacts—are how a research community learns which of its instruments actually measure what they claim. The audit code, data, preregistration, and one-command reproduction package are publicly archived.

Declaration of Generative AI Software tools in the writing process

During the preparation of this work, the author used Muse Spark (version 1.3, Meta) in all sections for drafting, analysis, and code validation, and for grammar correction and proofreading throughout. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Competing Interests

The author declares no competing interests. No funding was received for this work.

Data and Code Availability

The locked preregistration (`PREREGISTRATION.md`) with its deviation log, the 300-episode trajectory file, the real-data comparison pool, all analysis scripts, the post-hoc positive control, and a one-command reproduction package (`bash reproduce.sh`) are archived as supplementary material. A public code repository will be released upon acceptance.

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A. PREREGISTERED STATISTICAL TESTS: FULL RESULTS

All 24 two-sample Kolmogorov–Smirnov tests (simulated n=50 per archetype, minus preregistered exclusions, versus the 5,000-sequence real pool; Holm–Bonferroni, α=0.05) reject the null. The complete table with test statistics, raw and adjusted p-values, effective sample sizes, and bootstrap 95% confidence intervals for median differences is archived in `analysis/ks_tests.json` in the reproduction package.

B. REPRODUCTION PACKAGE CONTENTS

The reproduction package archives: the locked preregistration (`PREREGISTRATION.md`) with its deviation log; the 300-episode trajectory file (`results/trajectories.jsonl`); the real-data comparison pool and per-unit metrics; the analysis scripts that produced every table and figure; the post-hoc answer-injection positive control (`analysis/answer_injection_control.py` and `analysis/answer_injection_control.jsonl`); the strengthening analyses; and a verifier script that re-checks the headline numbers. One command (`bash reproduce.sh`) re-runs the full analysis from the banked artifacts.